

Stem-Cell Evidence: Benefit, Risk, Uncertainty

I can analyse simulated stem-cell evidence by calculating outcomes and identifying benefits, risks and limitations.

CALCULATE

frequency $\div$ total $\times$ 100

WEIGH

benefit and risk

LIMIT

what data cannot show

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What success looks like

- I can calculate an outcome percentage from a frequency.
- I can distinguish a possible benefit from a possible risk.
- I can identify why a small, short study leaves uncertainty.

Retrieve and reconnect

- Define benefit in this scientific context.
- Define risk without making it sound certain.
- Calculate $10 \div 20 \times 100$.
- Lead-in: why must a study's total number be reported?

In a fictional training study, 12 participants improved. Which missing fact is needed first?

Think first. Point, say, sign or write your choice. Give one reason.

- A. How many participants were in the group.
- B. The colour of the data table.
- C. Whether a headline called it a breakthrough.

Look closely · what is the evidence?

FREQUENCY · number observed

TOTAL · number studied

PERCENTAGE · $\text{frequency} \div \text{total} \times 100$

Look closely · what is the evidence?

group	improved	total
treated	12	20
control	6	20

Simulated results table

group	improved	total
treated	12	20
control	6	20

Totals highlighted

$$12 \div 20 \times 100 = 60\% \cdot 6 \div 20 \times 100 = 30\%$$

improvement recorded more often in the treated group — in this small, short, simulated study

Percentages

Words we will use

- benefit — possible helpful outcome
- risk — chance of harm
- uncertainty — what is not yet secure

Watch the model

- Label the table SIMULATED TRAINING DATA. Treatment: 12 of 20 improved; comparison: 6 of 20 improved.
- Calculate $12 \div 20 \times 100 = 60\%$ and $6 \div 20 \times 100 = 30\%$.
- Describe only what the table supports: improvement was recorded more often in the treatment group in this simulation.

Four of 20 participants experienced the defined adverse event. Which calculation gives the participant percentage?

A. $4 \div 20 \times 100 = 20\%$

B. $20 \div 4 \times 100 = 500\%$

C. 4% because four events occurred

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Add risk: four of 20 treatment participants and one of 20 comparison participants had the defined adverse event.
- Add uncertainty: each group is small, follow-up is short and the simulation gives no evidence about long-term outcomes.
- A balanced analysis states benefit evidence, risk evidence and at least one limit before any judgement.

Find and repair the misconception

Sixty per cent improved, so the simulated treatment is proven safe and effective.

- A. Keep it: one percentage proves both benefit and safety.
- B. Repair it: report the comparison, adverse events and study limitations.
- C. Repair it: percentages should never be used in Biology.

Benefit-risk-uncertainty analysis lab

Place each evidence statement, then use the frozen set to make one cautious analysis.

- READ DATA
- CALCULATE
- CLASSIFY
- LIMIT
- ANALYSE

Benefit-risk-uncertainty analysis lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- BENEFIT EVIDENCE
- RISK EVIDENCE
- UNCERTAINTY / LIMIT

What benefit, risk and uncertainty must appear in a scientifically cautious analysis?

Your independent mission

Analyse the simulated table: calculate four percentages, classify benefit and risk, and state two uncertainties.

Record: Visible percentage working plus a benefit-risk-uncertainty analysis that does not overclaim.

Choose your support and challenge

- Supported: Complete two percentage calculations with the formula shown, then match one benefit, risk and limit.
- Standard: Calculate all four percentages and write a three-part analysis.
- Stretch: Calculate all outcomes, compare groups and evaluate how sample size and follow-up affect confidence.

Show what you know

State one benefit finding and one uncertainty from the simulated data.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.